Exp No: 12 End-End Communication at Transport Layer.

8.10.26

Aim:-

To implement echo client server using TCP/UDP sockets program

Code:-

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower()=="quit":
            print("[SERVER] connection closed by client")
            break
        print(f"[CLIENT] Received: {data}")
        conn.send(f"Server received: {data}".
                  encode())
    conn.close()
    server.close()

def start_client():
    client = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] connected")
```

output:-

[server] Listening on 127.0.0.1:5000
[client] connected to server
[server] connected by ('127.0.0.1', [illegible])

Hello
(SERVER) Received: Hello
[SERVER RESPONSE] server received: Hello

quit
[SERVER] connection closed by client

UDP:

```
import socket
import threading

#--SERVER--

def udp_server():
    s = socket.socket(socket.AF_INET,
                      socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
    s.close()
    print("[SERVER] Stopped.")

#--CLIENT--
```

```
def udp_client():
    c=socket.socket(socket.AF_INET, SOCK_DGRAM)
    server=('127.0.0.1',12000)
    while True:
        msg=input("Enter message")
        c.sendto(msg.encode(), server)
        c.close()
if __name__=="__main__":
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```

Output :-

```
[SERVER] Running on 127.0.0.1:12000
Hi
Echo from server: Hi
Enter message (or 'quit' to exit): quit
[CLIENT] stopped. [SERVER] stopped.
```

Result :-

Thus the program to implement echo client server using TCP / UDP socket was executed successfully.